

1.1 Summary

This document defines system actors, system boundaries, functional requirements, and constraints based on the finalized use case diagram. It serves as the baseline document for further design activities such as UML diagrams and system implementation.

2. Introduction

2.1 Need for the System

Manual and semi-digital ticket reservation processes often suffer from:

- Delays in booking and confirmation
- Human errors
- Inefficient cancellation and refund handling
- Poor coordination between customers, service providers, and administrators

The Online Ticket Reservation System provides a centralized platform to automate ticket booking, cancellation, refunds, schedule management, and administrative control, thereby improving efficiency and transparency.

2.2 System Overview

The system enables customers to book and cancel tickets, administrators to manage users and view daily bookings, service providers to manage schedules and confirm trips, and

customer support staff to assist users through a centralized platform.

2.3 System Scope

The system includes the following features:

- Ticket booking and cancellation
- Refund processing
- Daily booking reports
- User and schedule management
- Customer assistance and coordination

2.4 Interaction with External Systems

The system interacts with:

- Payment systems for refund processing
- Centralized databases for ticket, user, and schedule storage

3. DEFINITIONS

Term	Definition
Online Ticket Reservation System	A system for booking and managing travel tickets
Customer	End user who books tickets
Administrator	System administrator managing users and reports

Term	Definition
Service Provider	Operator responsible for schedules and trips
Customer Support	Staff providing assistance to users
Booking	Confirmed ticket reservation
Refund	Return of payment after ticket cancellation

4. User Requirements Definition

4.1 Actors and Responsibilities

4.1.1 Customer

- Book tickets
- Cancel booked tickets
- Request assistance

4.1.2 Administrator

- View daily bookings
- Manage system users
- Process refunds

4.1.3 Customer Support

- Provide assistance to customers
- Respond to booking and cancellation queries
- Track and update the status of user complaints

- Escalate technical or payment-related issues to the administrator

4.1.4 Service Provider

- Add and update travel schedules
- Confirm trips
- Manage seat allocation and availability for trips

4.2 Functional Requirements

Booking and Cancellation

- The system shall allow customers to search available trips by route and date.
- The system shall display real-time seat availability before booking.
- The system shall allow customers to book tickets.
- The system shall generate a unique booking ID for each confirmed booking.
- The system shall allow customers to cancel booked tickets.
- The system shall automatically update seat availability after booking or cancellation.

Refund Management

- The system shall allow administrators to process refunds after ticket cancellation.

- The system shall calculate refund amounts based on predefined cancellation rules.
- The system shall record refund transaction details for auditing purposes.

Reporting

- The system shall allow administrators to view daily booking reports.
- The system shall allow administrators to filter booking reports by date and service.
- The system shall store historical booking data for reporting and analysis.

User Management

- The system shall allow administrators to add, update, or remove users.
- The system shall assign roles such as customer, service provider, customer support, and administrator.
- The system shall restrict system access based on assigned roles.

Assistance

- The system shall allow customers to submit assistance requests.

- The system shall allow customer support staff to view and respond to assistance requests.
- The system shall track the status of assistance requests until resolution.

Schedule and Trip Management

- The system shall allow service providers to add new travel schedules.
- The system shall allow service providers to update existing schedules.
- The system shall allow service providers to confirm trips.
- The system shall notify relevant users about confirmed or updated trips.

4.3 Non-Functional Requirements

- The system shall be easy to use with minimal learning effort.
- The system shall provide fast response times for booking and cancellation operations.
- The system shall ensure secure handling of user and transaction data.
- The system shall be reliable and available during operational hours.

5. System Architecture

5.1 High-Level Architecture

The system follows a modular architecture consisting of:

- Booking Module
- Cancellation and Refund Module
- User Management Module
- Schedule Management Module
- Assistance and Support Module
- Reporting Module

5.2 Module Interaction

- Customers interact with booking and assistance modules.
- Administrators interact with user management, reporting, and refund modules.
- Service providers interact with schedule and trip confirmation modules.
- Customer support interacts with the assistance module.

6. System Requirements Specification

6.1 Functional System Requirements

- The system shall authenticate users based on assigned roles.
- The system shall store booking, cancellation, and refund details securely.

- The system shall generate daily booking reports.
- The system shall maintain consistency between bookings and refunds.

6.2 Non-Functional System Requirements

Performance

- The system shall support multiple concurrent users.
- The system shall process booking and cancellation requests within acceptable response time.

Security

- The system shall authenticate users using secure login mechanisms.
- The system shall enforce role-based access control.
- The system shall protect sensitive user and payment data.

Reliability

- The system shall prevent double booking of seats.
- The system shall ensure data consistency during failures.

Usability

- The system shall provide a simple and intuitive user interface.
- The system shall display clear error messages and confirmations.

Scalability

- The system shall support growth in users and services.
- The architecture shall allow future feature expansion.

6.3 External Interfaces

- Web-based user interface
- Database interface for persistent storage
- Payment and refund interface

7. System Models

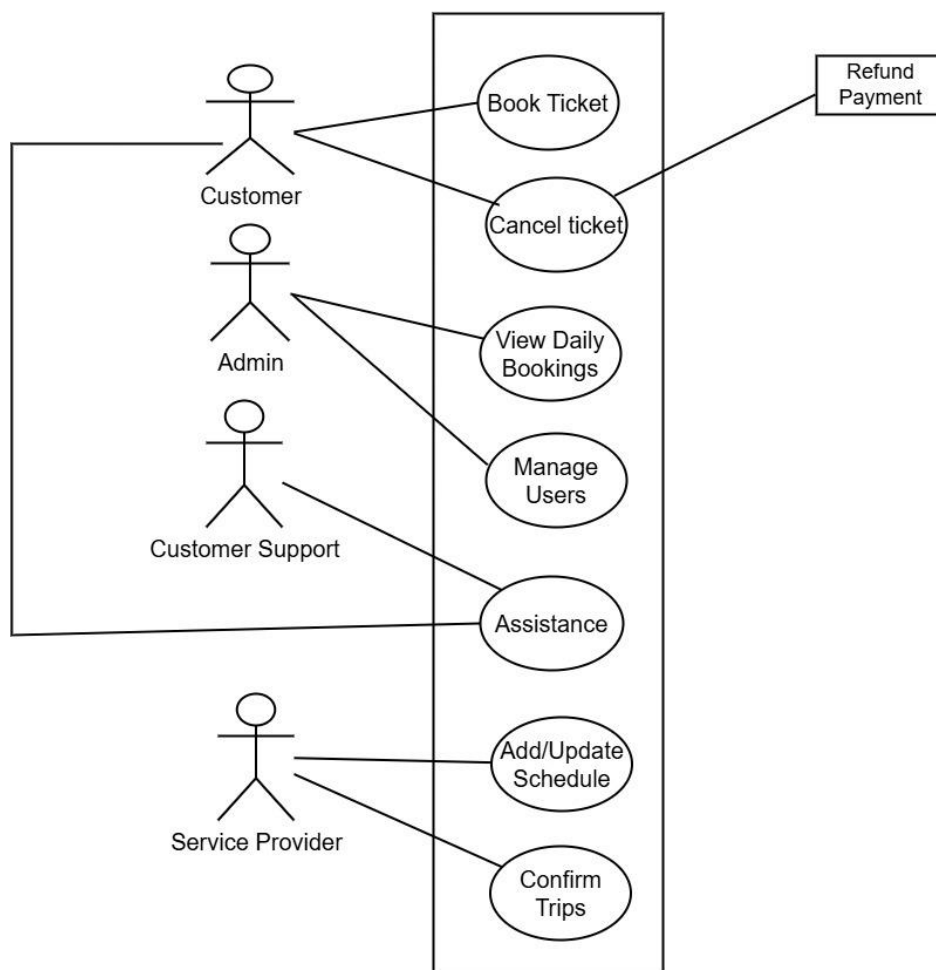
The system is represented using:

- Use Case Diagram
- Activity Diagrams (Booking, Cancellation)
- Sequence Diagrams (Booking, Refund)

USE CASE DIAGRAM

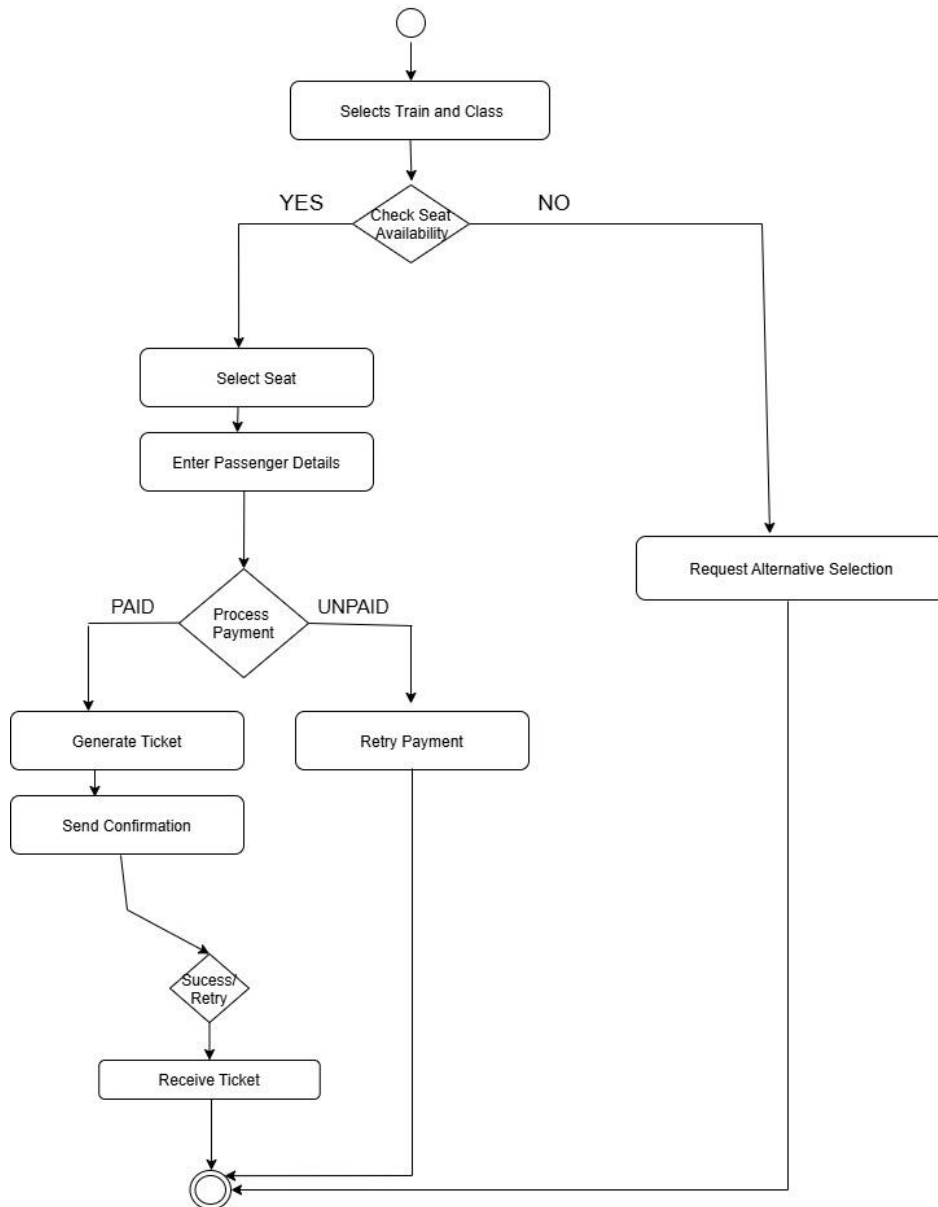
Online Ticket Reservation System

USE CASE

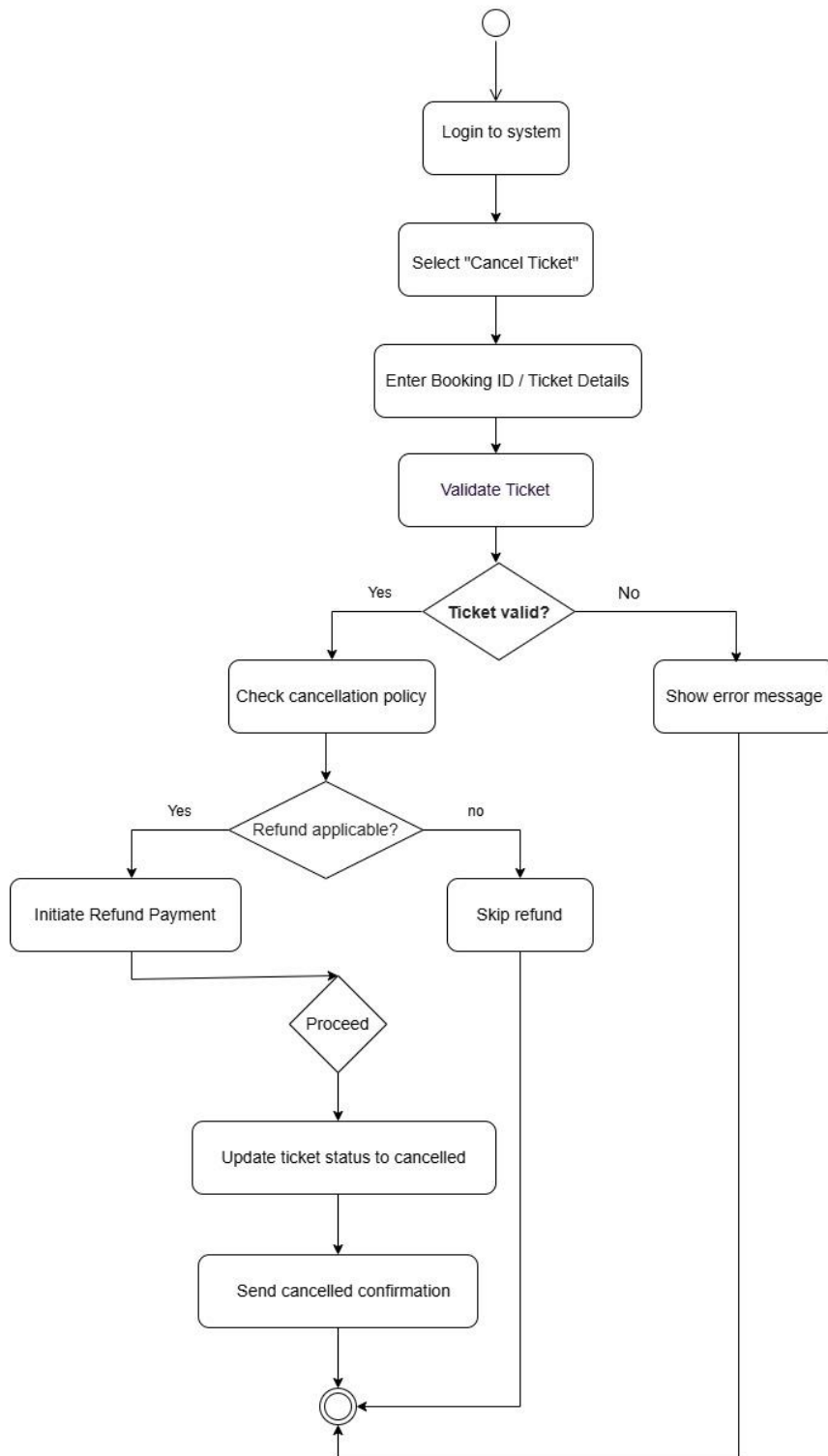


ACTIVITY DIAGRAMS

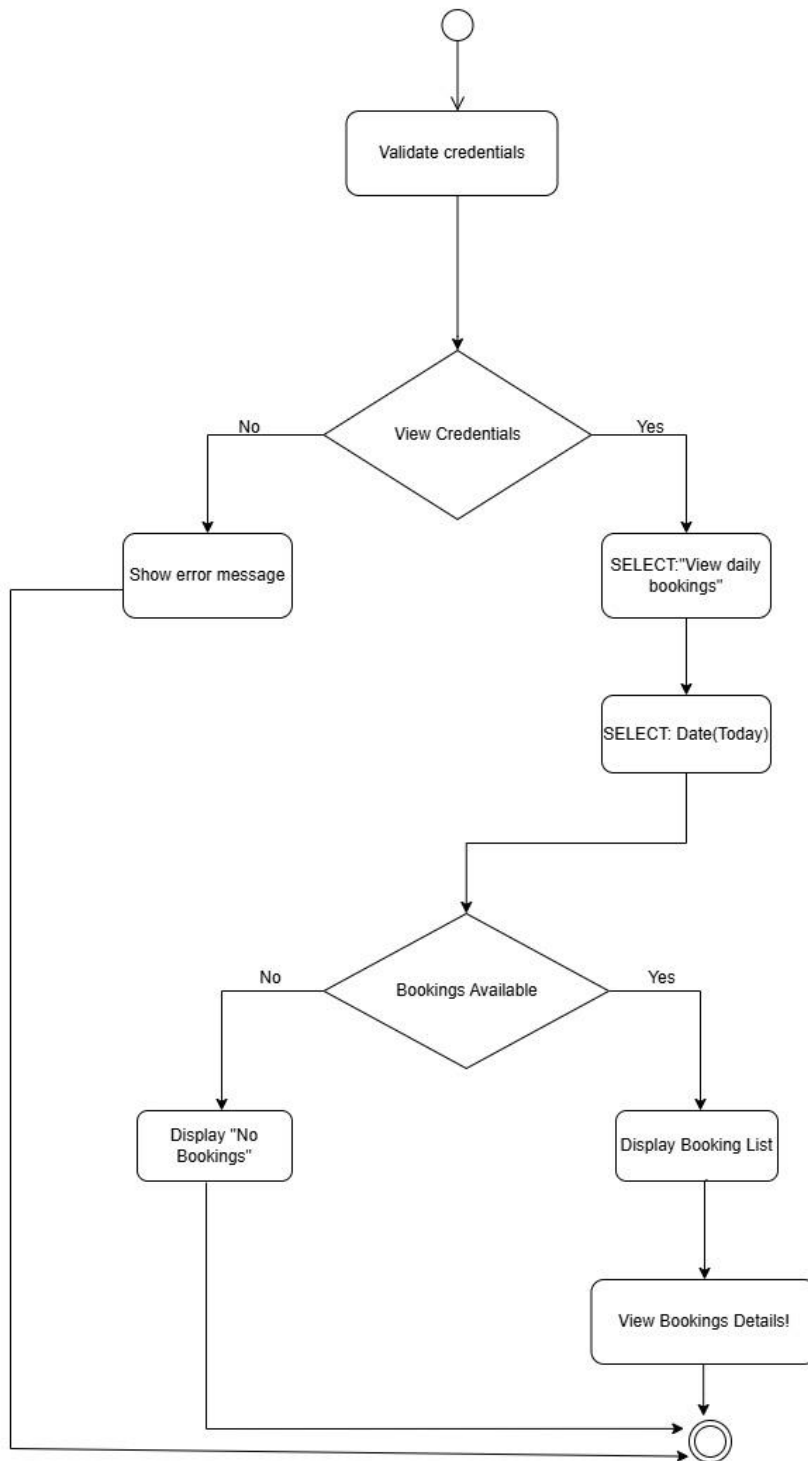
Book Ticket



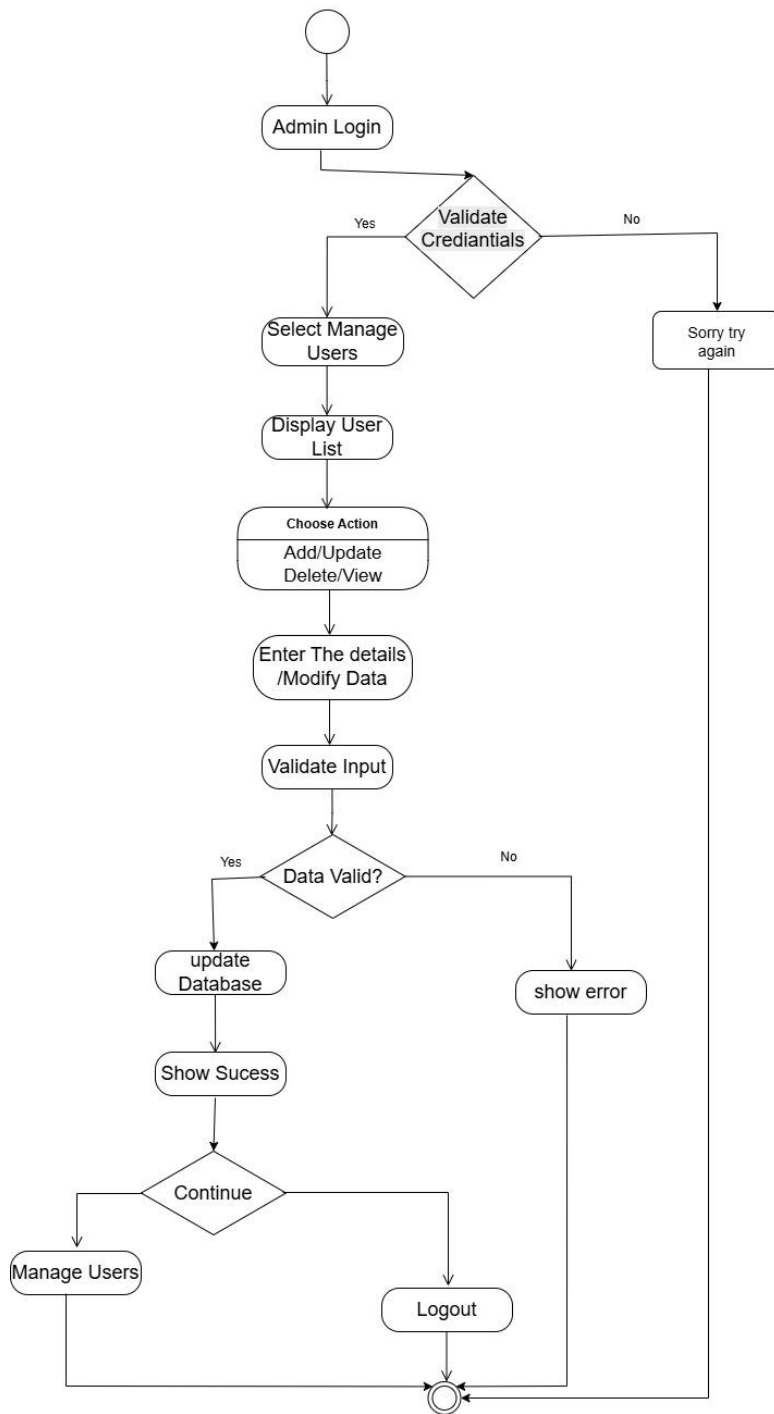
CANCEL TICKET



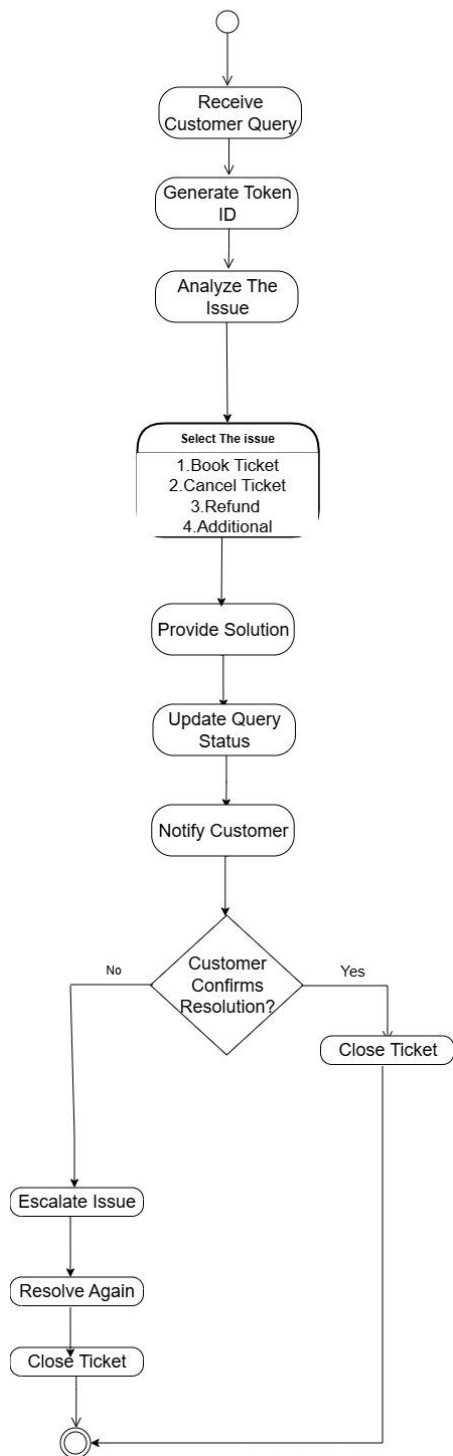
VIEW DAILY BOOKINGS



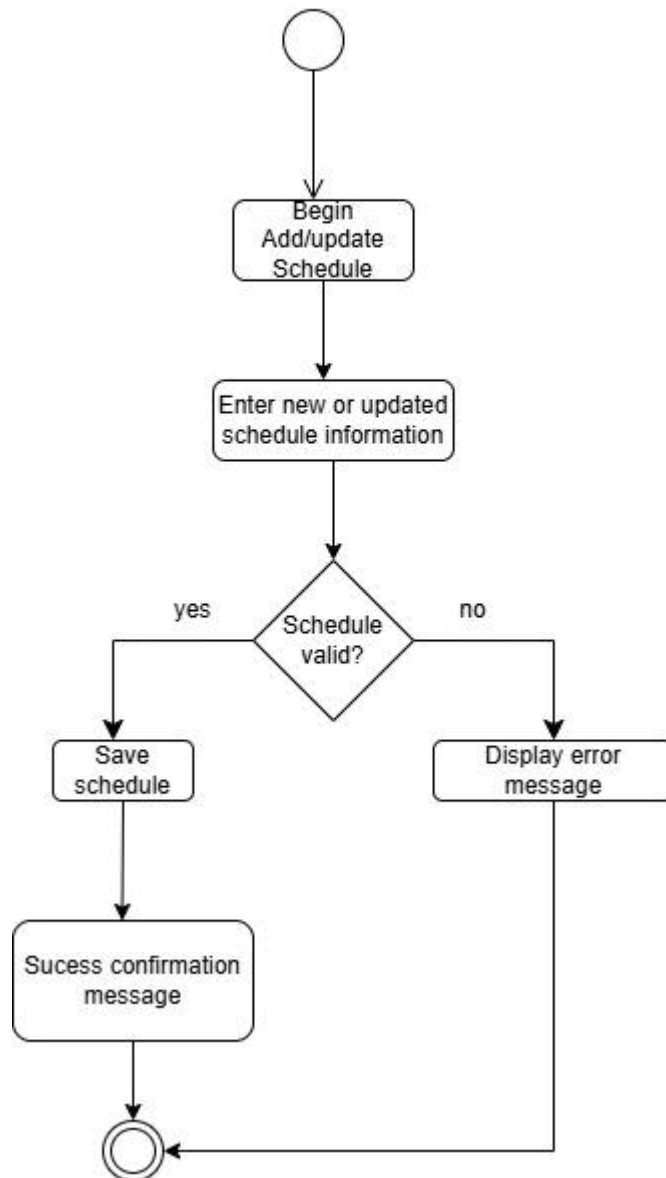
MANAGE USERS



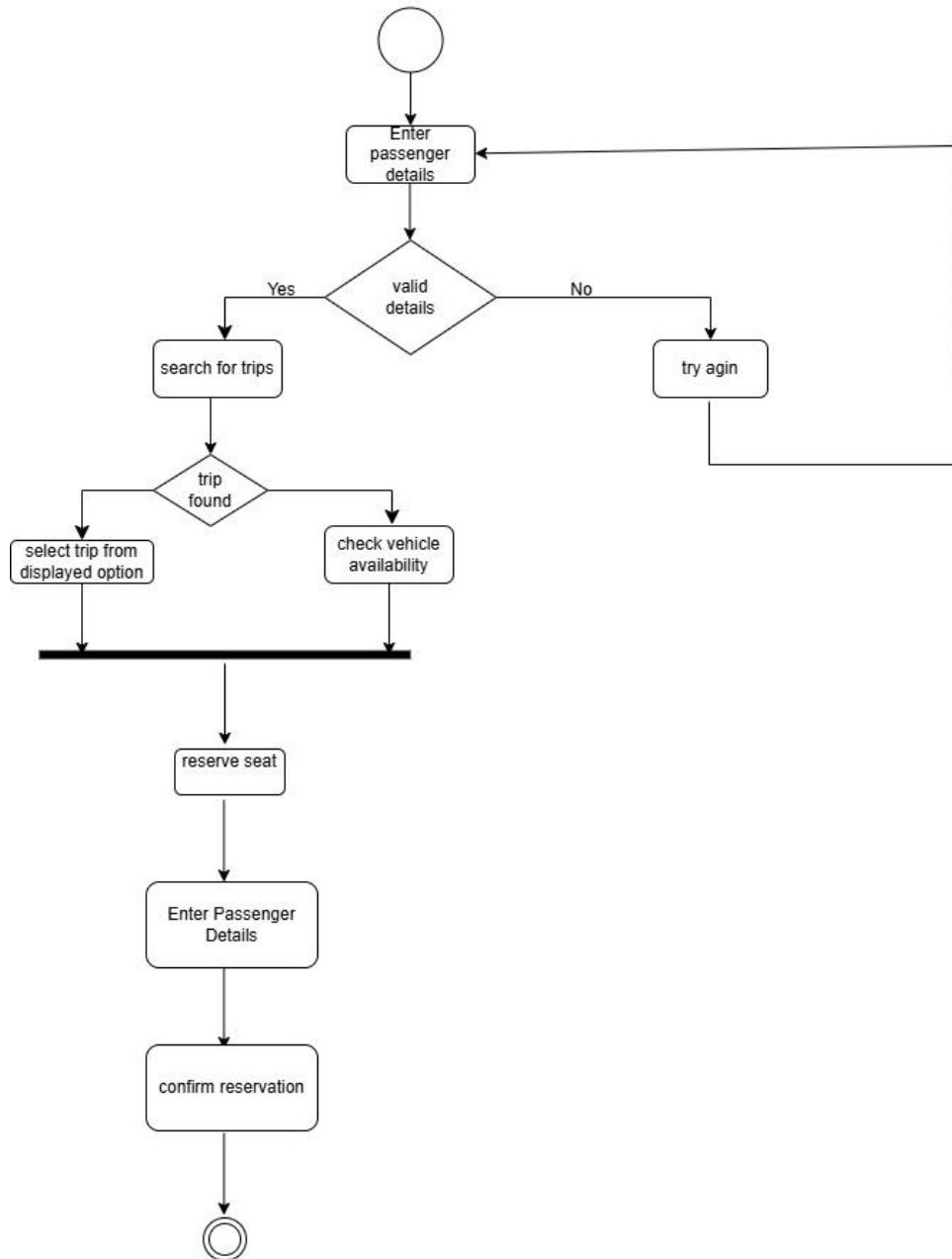
ASSISTANCE

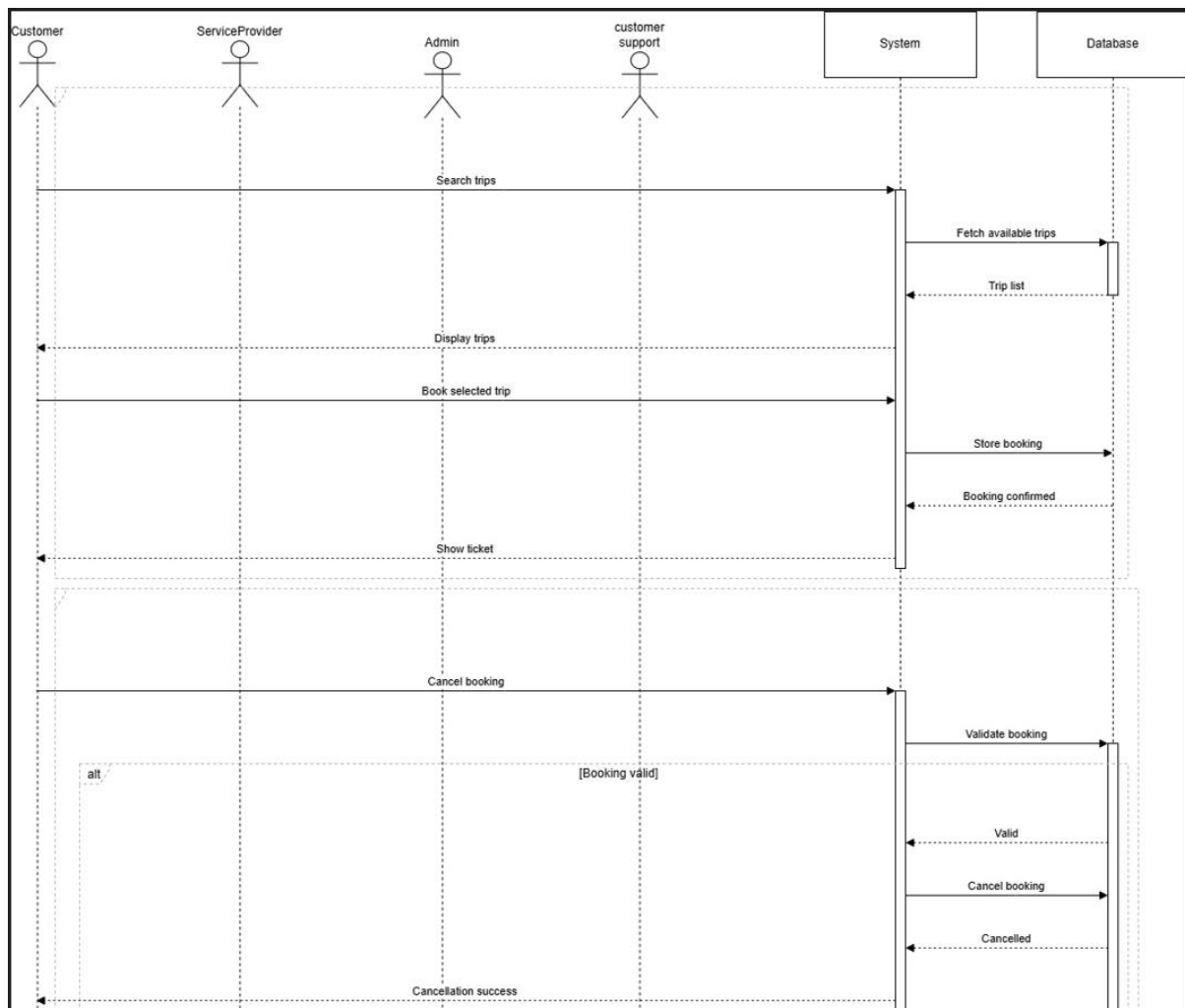


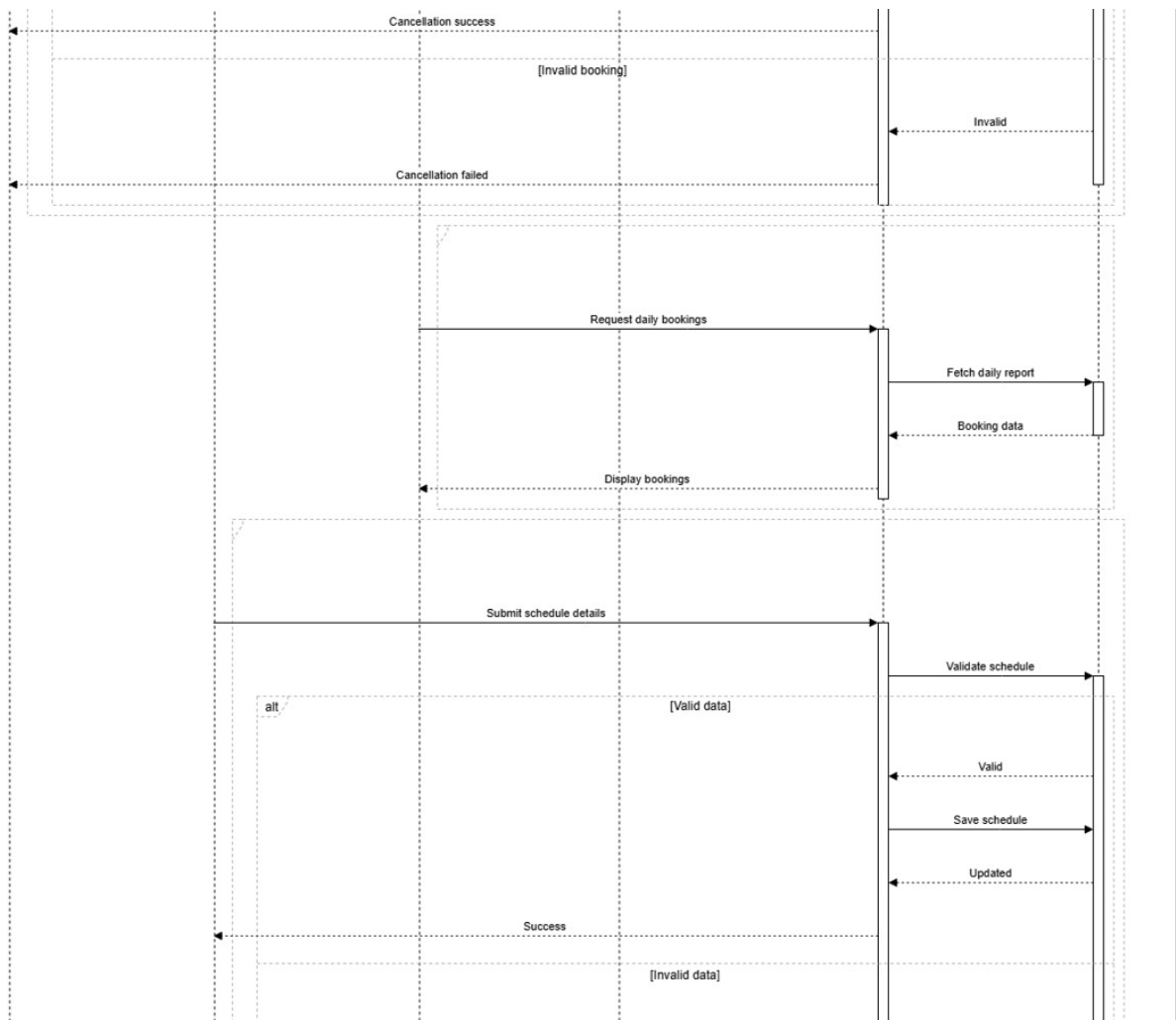
ADD / UPDATE

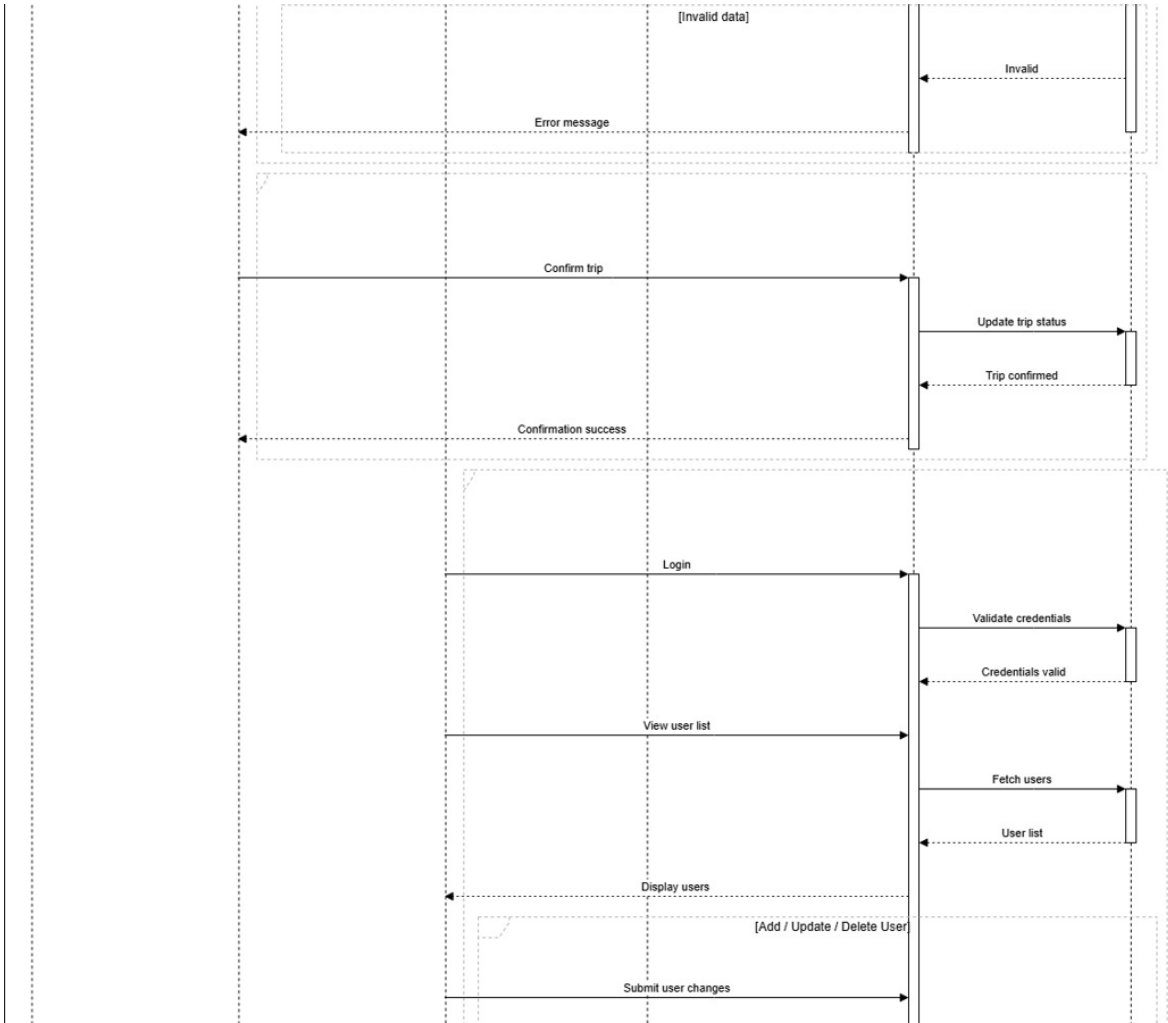


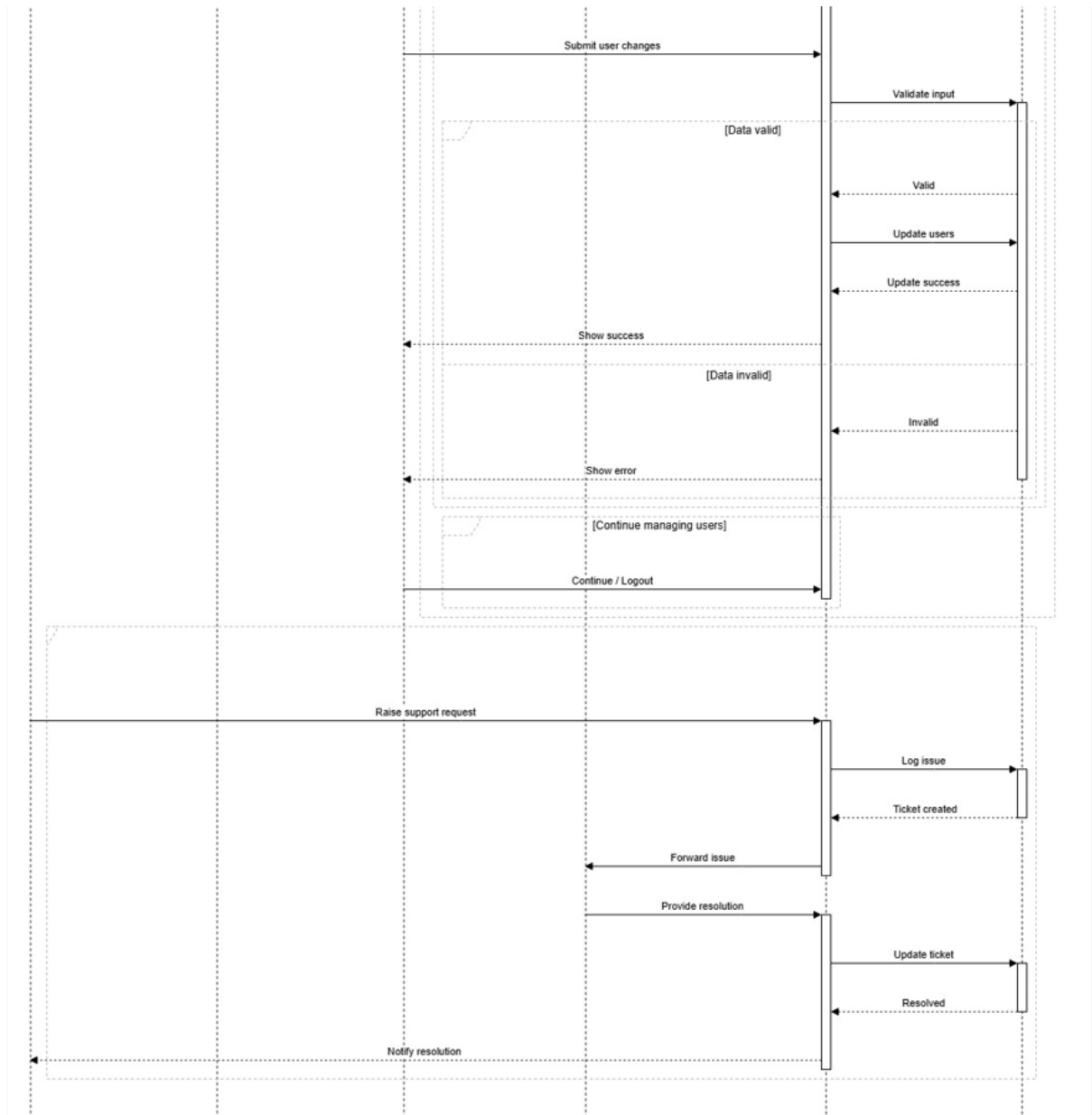
Confirm trip











8. System Evolution

8.1 Assumptions

- The system is accessed through a web application.
- Users have internet connectivity.
- Roles are predefined by administrators.

8.2 Future Enhancements

- Mobile application integration
- Online payment gateway integration
- Automated refund processing
- Advanced analytics and reporting

9. Appendices

9.1 Hardware Requirements

- Server with basic processing capacity
- Internet connectivity

9.2 Software Requirements

- Web browser
- Database management system

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